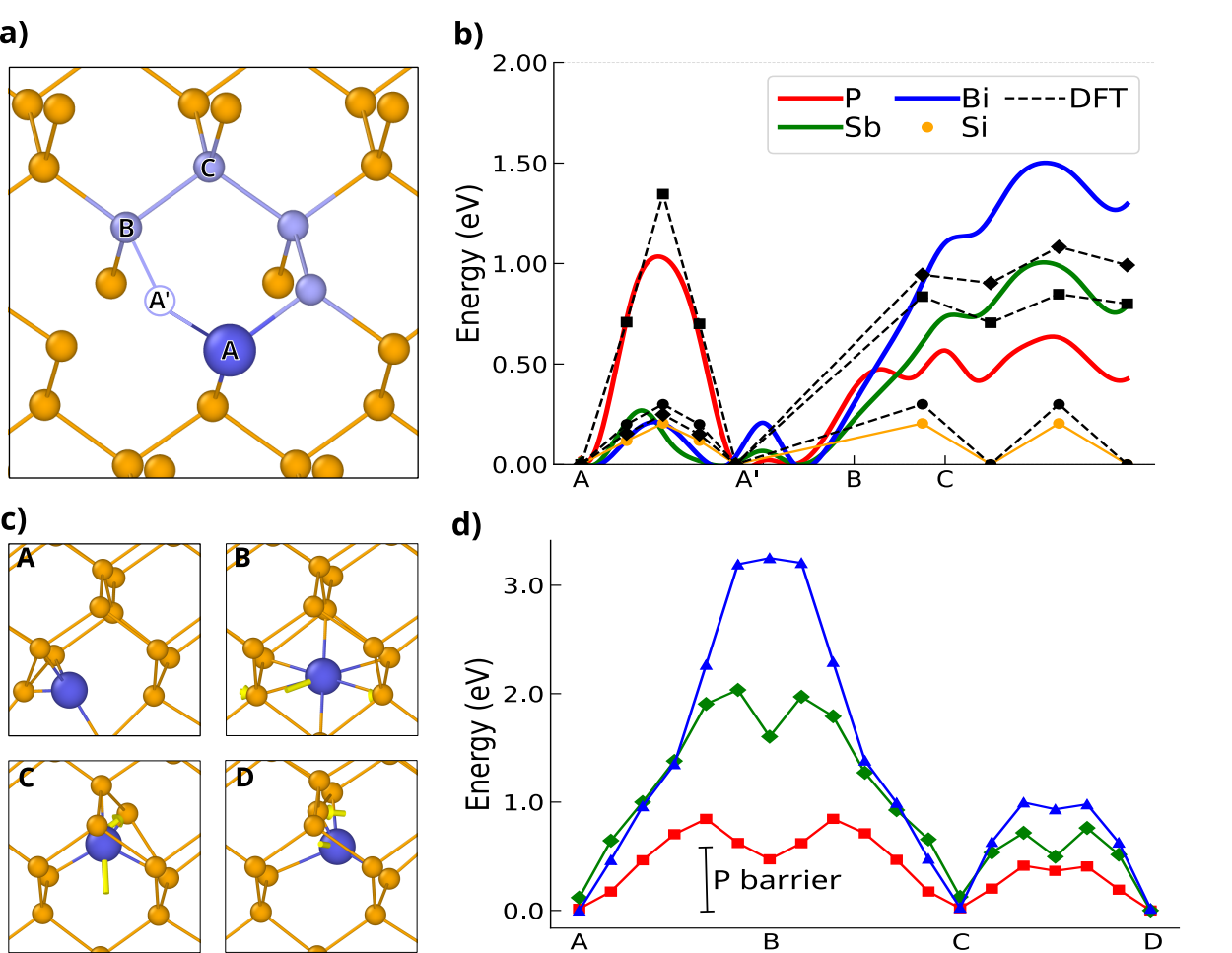

a)
C
B
A'
A
b)
2.00
1.50
1.00
0.50
0.00
Energy (eV)
P
Bi
DFT
Sb
Si
A
A'
B
C
c)
A
B
C
D
d)
3.0
2.0
1.0
0.0
Energy (eV)
P barrier
A
B
C
D